Hsuan-Fu (Jeff) Hsu

Embedded Systems and AI Engineering | C++, Python, Raspberry Pi, Arduino, AWS

Los Angeles, CA | +1 602-520-6877 | hsuanfu0709@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science — Sept 2020 - June 2024

SKILLS

Proficient: C++, Python, C#, Git

Familiar: Raspberry Pi 4, Arduino Nano 33 BLE Sense, Arm Cortex-M4, I2C, UART, BLE, Renode, GDB, Linux, OpenGL, GLSL, AWS (Lambda, SQS, DynamoDB, CDK), Docker, pytest, GitHub Actions, SQL, TypeScript, PyTorch

Exposure: TensorFlow, tsfresh

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Co-developed a portable NIR sensing device with two teammates, using an **Arduino Nano 33 BLE Sense** and an AS7263 sensor connected over I2C. The device sampled 810/860 nm signals at 34 Hz and transmitted readings over BLE.
- Built a **Python** pipeline to process NIR signals from 30 participants. The team's best model estimated uric acid levels with 10.3% MAPE in leave-one-subject-out testing.
- Received an individual NSTC research grant and placed fourth in NTPU's project competition as part of a three person team.

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Built a birth grant application system using **ASP.NET Core MVC** and **SQL Server** as the sole developer, and deployed it to 29 government offices.
- Optimized SQL Server processing and storage, reducing database latency by at least 50% and resource usage by at least 30%.

PROJECTS

Cortex-M4 Emulator and Bare-Metal Firmware — Nov 2026 - Present
C, Arm Thumb-2, STM32F446, Renode, GDB | GitHub

- A **Cortex-M4** emulator in C that runs bare-metal STM32F446 firmware with no board and models six chip peripherals.
- Compared it with **Renode** one instruction at a time. 33 of 35 test programs matched, and so did the first 16,439 firmware instructions.
- The no-HAL firmware drives UART, a BME280, and an SSD1306 and passes seven Renode tests. It has not run on a board yet.

Classroom AC Control System — USC IEEE HackIoT (team of 4) — Oct 2026
Raspberry Pi 4, Python, YOLO, DHT11, Supabase, TypeScript | GitHub

- Set up a **Raspberry Pi 4** with teammates to count classroom occupants using YOLO at approximately 7 FPS and log DHT11 temperature and humidity readings to Supabase every 30 seconds.
- Ported a Python air-conditioning model to **TypeScript** and verified matching results across eight weeks of simulated operation. Built a Next.js simulator that estimated 27% energy savings for one room over that period.

3D Rendering Engine — Oct 2022 - Jan 2023; rebuilt Nov 2026
C++, OpenGL, GLSL, CMake | GitHub

- Rebuilt a **C++/OpenGL** renderer with an OBJ/MTL parser and vertex deduplication, reducing a test model's vertex count by 71% and parse time from 108 ms to 56 ms.
- Added Cook-Torrance PBR with image based lighting, PCF shadows, and glTF 2.0 loading. Fuzz tested both loaders under ASan/UBSan, with no issues found across more than 710,000 OBJ inputs.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, 2026, doi: 10.1109/THMS.2026.3724413.